THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 10 — MATHS · HARD LEVEL UNIT TEST SET 1

Chapters 12–14: **Surface Areas and Volumes, Statistics, Probability** | RBSE Pattern | Session 2026–27

Time: 2 Hours Name: _____ Roll No.: _____ **Maximum Marks: 50**

General Instructions:

1. All questions are compulsory.
2. Q. 1–15 are Objective Type (1 mark each).
3. Q. 16–20 are Very Short Answer Type (2 marks each).
4. Q. 21–25 are Short Answer Type (3 marks each).
5. Q. 26–27 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 24–25 and Q. 26–27.
7. Write the question number before each answer.
8. Hard level: application and higher-order questions.

SECTION — A (Objective Type Questions) 15 × 1 = 15

Choose the correct option and write it in your answer book.

1. The largest possible solid sphere is carved out of a cube of edge 21 cm. The volume of the material wasted is: 1

(a) 4410 cm³ **(b)** 6835.5 cm³ **(c)** 4851 cm³ **(d)** 9261 cm³

2. The mean of a distribution with classes 0–10, 10–20, 20–30, 30–40, 40–50, 50–60, 60–70, 70–80 and frequencies 3, 4, p, 4, 15, 10, 19, 22 respectively is 49. The value of p is: 1

(a) 22 **(b)** 25 **(c)** 23 **(d)** 28

3. Two fair dice are thrown together and the numbers on the upper faces are noted. The probability that the sum of the numbers is 8 is: 1

(a) $\frac{1}{9}$ **(b)** $\frac{1}{6}$ **(c)** $\frac{31}{36}$ **(d)** $\frac{5}{36}$

4. A solid sphere of radius 10.5 cm fits exactly inside a cylinder (it touches the curved surface and both bases). The volume of the space in the cylinder not occupied by the sphere is: 1

(a) 2425.5 cm³ **(b)** 12127.5 cm³ **(c)** 7276.5 cm³ **(d)** 4851 cm³

5. The median of a distribution with classes 200–210, 210–220, 220–230, 230–240, 240–250, 250–260, 260–270 and frequencies 12, 8, x, 25, y, 18, 19 respectively, and total frequency 100, is 238.4. The value of x is: 1

(a) 10 **(b)** 8 **(c)** 11 **(d)** 9

6. Two fair dice are thrown together and the numbers on the upper faces are noted. The probability that the sum is less than 5 is: 1

(a) $\frac{5}{36}$ **(b)** $\frac{5}{6}$ **(c)** $\frac{7}{36}$ **(d)** $\frac{1}{6}$

7. A solid cylinder of radius 6 cm and height 5 cm is melted and recast into a cone of the same base radius. The height of the cone is: 1

(a) 30 cm **(b)** 15 cm **(c)** 10 cm **(d)** 5 cm

8. For the data with classes 200–220, 220–240, 240–260, 260–280, 280–300, 300–320, 320–340 with frequencies 23, 4, 25, 6, 3, 19, 20 respectively, The mean of the data is: 1

(a) 269.8 **(b)** 270.8 **(c)** 268.8 **(d)** 270

9. A bag contains 15 red balls and x blue balls. The probability of drawing a blue ball is 3 times the probability of drawing a red ball. The value of x is: 1

(a) 60 **(b)** 45 **(c)** 44 **(d)** 3

10. A solid cone and a solid hemisphere have equal base radii 7 cm and equal volumes. The height of the cone is: 1

(a) 10.5 cm **(b)** 21 cm **(c)** 14 cm **(d)** 7 cm

11. For the data with classes 20–40, 40–60, 60–80, 80–100, 100–120, 120–140, 140–160 with frequencies 3, 13, 11, 17, 15, 24, 17 respectively, The median of the data is: 1

(a) 120 **(b)** 103.6 **(c)** 108 **(d)** 100

12. A bag contains 7 red and 9 blue balls. When x more red balls are added, the probability of drawing a red ball becomes $\frac{4}{7}$. The value of x is: 1

(a) 4 **(b)** 10 **(c)** 6 **(d)** 5

13. **Assertion (A):** The total surface area of a solid hemisphere of radius r is $3\pi r^2$. 1

Reason (R): The total surface area of a solid hemisphere is its curved surface area $2\pi r^2$ plus the area πr^2 of its flat circular face.

Choose the correct option:

(a) Both A and R are true and R is the correct explanation of A.

(b) Both A and R are true but R is not the correct explanation of A.

(c) A is true but R is false.

(d) A is false but R is true.

14. **Assertion (A):** For the class interval 10-20, the class mark is 15. 1

Reason (R): The class mark of a class interval is $\frac{\text{upper limit} + \text{lower limit}}{2}$.

Choose the correct option:

(a) Both A and R are true and R is the correct explanation of A.

(b) Both A and R are true but R is not the correct explanation of A.

(c) A is true but R is false.

(d) A is false but R is true.

15. **Assertion (A):** The probability of getting a number greater than 6 when a die is thrown once is 0. 1

Reason (R): The probability of an impossible event is 0.

Choose the correct option:

(a) Both A and R are true and R is the correct explanation of A.

(b) Both A and R are true but R is not the correct explanation of A.

(c) A is true but R is false.

(d) A is false but R is true.

SECTION — B (Very Short Answer Type) 5 × 2 = 10

Answer in about 20–30 words each.

16. A solid sphere of radius 6 cm is completely dropped into water in a cylindrical vessel of base radius 8 cm, and no water spills. Find the rise in the water level. 2

17. The mean of a distribution with classes 200–220, 220–240, 240–260, 260–280, 280–300, 300–320, 320–340 and frequencies 3, 24, p, 10, 15, 14, 17 respectively is 274. Find the value of p. 2

18. Two fair dice are thrown together and the numbers on the upper faces are noted. Find the probability that the sum is a multiple of 4. 2

19. A hemispherical bowl has outer radius 7.5 cm and inner radius 4.5 cm. Find the volume of material used to make the bowl. 2

20. The marks obtained by students in a test are given below. 2

Marks	0–10	10–20	20–30	30–40	40–50	50–60	60–70	70–80
Number of students	3	4	23	4	15	10	19	22

Find the mode.

SECTION — C (Short Answer Type) — 5 × 3 = 15

Answer in about 40–60 words each.

21. A solid metallic cylinder of base radius 9 cm and height 32 cm is melted and recast into solid spheres of radius 6 cm. Find the number of spheres obtained. 3

22. The marks obtained by students in a test are given below. 3

Marks	40–50	50–60	60–70	70–80	80–90	90–100	100–110	110–120
Number of students	11	19	8	20	23	2	13	4

Find the mean and the mode (in this order).

23. Two fair dice are thrown together and the numbers on the upper faces are noted. Find the probability that the sum is 9. 3

24. A solid metallic cone of base radius 10 cm and height 5 cm is melted and recast into a solid sphere. Find the radius of the sphere. 3

OR

The marks obtained by students in a test are given below. 3

Marks	40–50	50–60	60–70	70–80	80–90	90–100	100–110
Number of students	17	21	24	17	6	4	11

Find the mean.

25. A bag contains 9 red and 7 blue balls. When x more red balls are added, the probability of drawing a red ball becomes $\frac{2}{3}$. Find the value of x. 3

OR

A solid cylinder of base radius 6 cm and height 20 cm is melted and recast into identical solid cones, each of base radius 3 cm and height 8 cm. Find the number of cones obtained. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

26. **Case study.** A medicine capsule is in the shape of a cylinder with a hemisphere stuck at each end. The radius of the cylinder and of each hemisphere is 3.5 mm and the length of the cylindrical part is 7 mm. (Take $\pi = \frac{22}{7}$.) 5

(i) Find the curved surface area of the cylindrical part.
(ii) Find the surface area of the two hemispherical ends together.
(iii) Find the total surface area of the capsule.
(iv) Find the total length of the capsule and the volume of the cylindrical part.

27. Two dice are thrown together. Find the probabilities of these events (in this order): 5

(i) the product of the numbers is 12,
(ii) the product is a perfect square,
(iii) the first number is greater than the second.

OR

A solid sphere of radius 31.5 cm fits exactly inside a cylinder (it touches the curved surface and both bases). Find the volume of the space in the cylinder not occupied by the sphere. 5

END OF QUESTION PAPER — **Total: 50 Marks**